Does the geometric difference predict when a quantum kernel helps?

A test of four hypotheses on ATLAS $H \to \tau\tau$ events

qkernel-hep project

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Abstract

We test whether the geometric difference g of Huang et al. predicts how fidelity and projected quantum kernels compare with tuned classical models on 818,238 simulated ATLAS $H \to \tau\tau$ events, with 500 training events, 5 seeds, 2–14 qubits, Z and ZZ feature maps and a bandwidth sweep (79 configurations). Four hypotheses were written before the experiments. **H1 held**: no quantum kernel beats the best classical baseline by more than the seed spread (0 of 79 configurations, for both kernels). **H2 failed**: $\rho(g, \text{gap}) = -0.44$ $[-0.61, -0.22]$ for the regularised g ($\lambda = 0.01$), the wrong sign. This is consistent with Huang et al.'s necessary condition: the regularised g never exceeds $\sqrt{N} = 22.4$ (maximum 19.6). What fails is the graded use of g to pre-screen datasets. Kernel concentration explains the results better. **H3 held in part** (4 of 15 cells). **H4 held in part**: a smaller bandwidth recovers AUC in 15 of 15 cells, but g does not return towards 1. Boosted-tree feature selection instead of PCA changes no verdict. On `ibm_marrakesh` the AUC is 0.681 against 0.722 exact (24 test events). We find no quantum advantage.

1 Introduction

Quantum kernel methods embed data x in a quantum state $|\phi(x)\rangle$ and train a classical kernel machine on the overlaps of these states. Whether they can beat classical models on classical data is open. Huang et al. [13] proposed a data-dependent test: the *geometric difference* $g(K_C \,\|\, K_Q)$ bounds how much harder a classical model can find the labels than the quantum model. If $g \lesssim \sqrt{N}$, a classical kernel is guaranteed to be competitive. The test has since been used as a *pre-screening* tool, with a larger g read as more promise for the quantum kernel [15, 17].

We ask whether this graded reading holds on collider data. We use the ATLAS Higgs Machine Learning Challenge events [1, 3], a standard signal-versus-background benchmark. Quantum kernels have been applied to collider data before [4, 5, 8, 10, 12, 18, 24], including to this dataset [6], but none of these studies tests g. Outside physics, g has been reported not to track performance on tabular, financial, power-system and materials data [2, 9, 14, 20]. In parallel, kernel concentration [23] and bandwidth tuning [7, 11, 19, 21] have emerged as the factors that decide whether a quantum kernel learns at all.

Our contribution is a test of g and of the label-aware criterion on collider data, against four hypotheses written before the experiments (in the project guide, 7 November 2026; not formally pre-registered). The tests use seed-paired significance, bootstrap confidence intervals, correlations within each feature-map family and with concentration held fixed, and a replication with a second feature reduction. The results are *consistent* with Huang et al.'s necessary condition. What fails is the graded use of g to pre-screen datasets. Kernel concentration explains the pattern better.

2 Background

Fidelity and projected kernels. For a data-encoding unitary $U(x)$, the fidelity kernel is $K_Q(x, x') = |\langle 0|U(x')^\dagger U(x)|0\rangle|^2$. The projected kernel [13] instead compares single-qubit reduced density matrices $\rho_k(x)$: $K_P(x, x') = \exp\big(-\gamma \sum_k \|\rho_k(x) - \rho_k(x')\|_F^2\big)$. We use Qiskit's Z and ZZ feature maps with linear entanglement and r repetitions. Each repetition applies Hadamards and then a diagonal phase: $2x_i$ on qubit i, and for the ZZ map $2(\pi - x_i)(\pi - x_j)$ on neighbouring pairs. The bandwidth c rescales the inputs, $x \to c\,x$ [19].

Geometric difference and model complexity. With both kernels normalised to trace N, Huang et al. define $g(K_C \,\|\, K_Q) = \sqrt{\|\sqrt{K_Q} K_C^{-1} \sqrt{K_Q}\|_\infty}$, where $\|\cdot\|_\infty$ is the spectral norm. A regularised version replaces K_C^{-1} by $K_C(K_C + \lambda I)^{-2}$ [13]. In their argument, g bounds the classical model complexity: $s_C \le g^2 s_Q$. A

quantum advantage is therefore only possible if $g \gtrsim \sqrt{N}$ *and* the label-aware complexities satisfy $s_Q \ll s_C$. We take K_C to be the RBF kernel with the RBF-SVM's cross-validated γ. Our s is the ridge-regularised $\alpha^\top K\alpha/N$ with $\alpha = (K + \lambda I)^{-1}y$, which equals $s_K(N)/N$ in Huang et al.'s notation; their error bound $\sqrt{s_K(N)/N}$ is then $\sqrt{s}$.

Concentration and bandwidth. For expressive embeddings, global measurements, entanglement or noise, kernel values concentrate exponentially in the number of qubits around a fixed value [23]. The Gram matrix then approaches a constant plus the identity, and the model stops depending on its inputs. Shrinking c counteracts this, and an optimally tuned quantum kernel tends to behave like a classical RBF kernel [7, 11, 19, 21]. We measure concentration by the variance of the off-diagonal training-kernel entries.

3 Data and preprocessing

The dataset has 818,238 simulated events (279,560 signal, 538,678 background) with 30 features. We keep the 19 features that are always defined (no -999 placeholder), replace raw azimuthal angles by the rotation-invariant differences $|\Delta\phi|$ between the tau, lepton and missing momentum, and draw disjoint, class-balanced samples of 500 training and 1,000 test events per seed. A StandardScaler and PCA to n components (n = number of qubits) are fitted on the training events only. Each component is mapped linearly from its training range onto $[-\pi, \pi]$ and then multiplied by c. As a robustness check (Sec. 6) PCA is replaced by the n features with the largest boosted-tree importance, again fitted on training events only.

4 Method

Grid. The main grid covers $n \in \{2, 4, 6, 8, 10\}$, three feature maps (Z with $r = 2$; ZZ with $r = 1$ and $r = 2$), $c \in \{0.05, 0.1, 0.25, 0.5, 1\}$ and 5 seeds: 375 runs. It is extended to $n \in \{12, 14\}$ for ZZ, $r = 2$ with $c \in \{0.1, 1\}$ (20 runs). Together these give the 79 configurations on which the hypotheses are judged. A training-size sweep ($100, 250, 500, 1000$ events, 80 runs) and the feature-selection replication (150 runs) are reported separately. Kernels are computed by exact statevector simulation, cross-checked against Qiskit to 10^{-10}.

Models and tuning. The quantum kernels feed a support vector machine (SVM) with a precomputed kernel, with C tuned by stratified 5-fold cross-validation (CV) on the training events (ROC-AUC). For the projected kernel, γ is also tuned by CV, relative to the median distance. The classical baselines are logistic regression, an RBF-SVM and boosted trees (XGBoost), each tuned by the same CV. The metric is test ROC-AUC. "Best classical" is the maximum of the three classical test AUCs, which favours the classical side (Sec. 8). g is computed for $\lambda \in \{0, 0.001, 0.01, 0.1\}$; the headline uses $\lambda = 0.01$.

Statistics. A seed-paired difference is *not significant* if its mean is within one seed-to-seed standard deviation. With 5 seeds this is a lenient rule: roughly a paired t of 2.2, $p \approx 0.09$. Correlations are Spearman's ρ over configuration means, with 95% percentile-bootstrap intervals over configurations (2,000 resamples). Partial correlations control for $\log_{10}$ of the off-diagonal variance.

5 Results

The hypotheses, quoted from the project guide:

> **H1:** The tuned classical baselines match or beat every quantum kernel on this data. Test: AUC with error bars across all configurations.
> **H2:** Geometric difference correlates positively with the quantum-minus-classical AUC gap. Test: Spearman correlation across configurations, with a confidence interval.
> **H3:** As qubits and depth grow at fixed bandwidth, concentration rises and AUC falls. Test: plot variance and AUC against qubit count.
> **H4:** Reducing bandwidth c reduces concentration and recovers performance, while pulling g back toward 1. Test: sweep c at fixed qubits; plot g, variance and AUC together.

5.1 H1: classical baselines versus quantum kernels

Across all 79 configurations, the fidelity kernel is significantly worse than the best classical baseline in 58 and not significantly different in 21. The projected kernel is worse in 60 and indistinguishable in 19. Neither

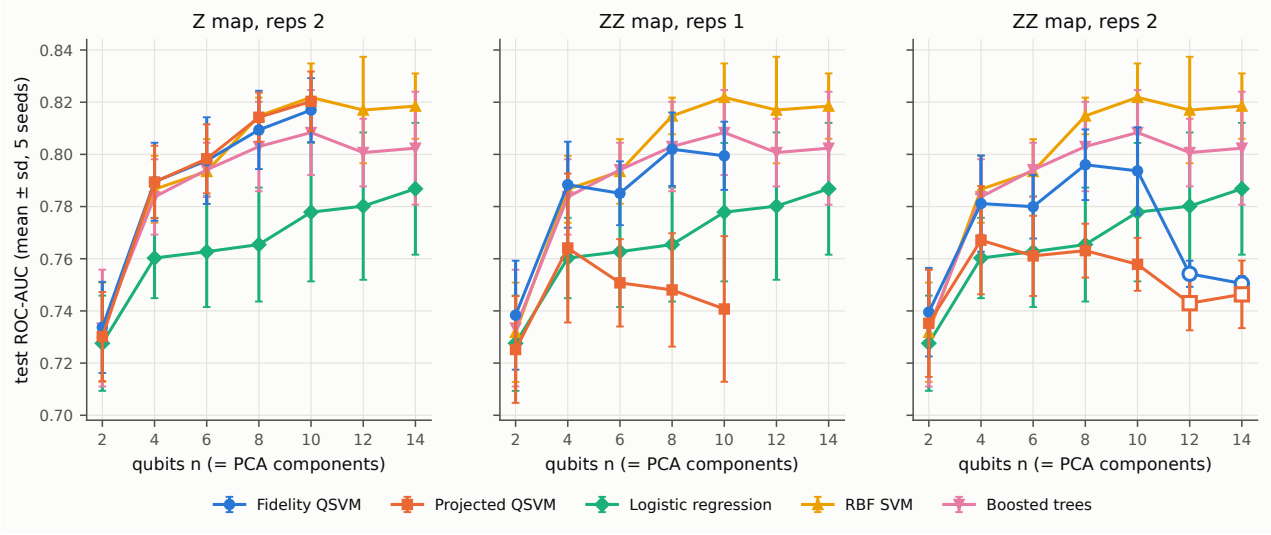


Figure 1: Test AUC against qubit count for every model (mean ± sd over seeds), one panel per feature map. The quantum bandwidth is chosen per seed by CV; open markers: only $c \in \{0.1, 1\}$ searched.

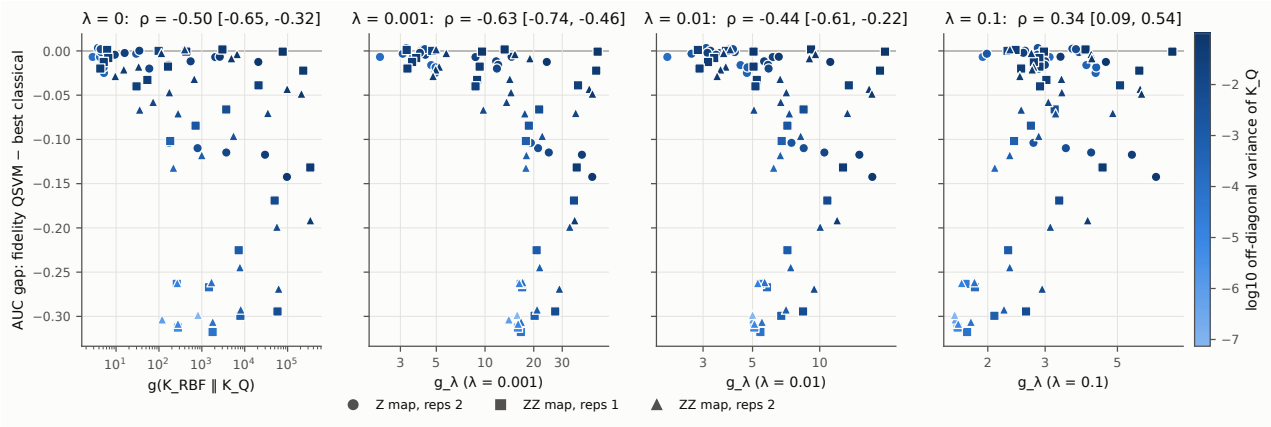


Figure 2: AUC gap (fidelity kernel minus best classical) against g for four λ, one point per configuration, coloured by $\log_{10}$ off-diagonal variance; Spearman ρ with 95% bootstrap interval above each panel.

kernel is significantly better anywhere (0 of 79), and the largest mean gap is +0.0033 (fidelity) and +0.0019 (projected). Against the RBF-SVM alone, 2 configurations exceed it by more than the seed spread: ZZ, reps 1, $n = 2$, $c = 0.1$ (+0.0071 ± 0.0058) and ZZ, reps 2, $n = 2$, $c = 0.05$ (+0.0073 ± 0.0038). Both are at the smallest n, and both lie within the seed spread of the best classical model (+0.0017, +0.0018). About 3.6 such hits per kernel are expected by chance under this rule. Figure 1 shows the per-seed best bandwidth. The best classical AUC rises from 0.737 at $n = 2$ to 0.822 at $n = 10$. The Z map tracks it (0.817 at $n = 10$, gap −0.0051). The ZZ maps fall significantly behind from $n = 6$ (gap −0.028 at $n = 10$ and −0.071 at $n = 14$ for $r = 2$). **Verdict: H1 held.**

5.2 H2: geometric difference versus the AUC gap

Table 1 and Fig. 2 give the test. For the headline regularised g ($\lambda = 0.01$), $\rho = -0.44$ [−0.61, −0.22]: significantly *negative*. The same holds at $\lambda = 0$ and 0.001. Only $\lambda = 0.1$ gives a positive pooled ρ (+0.34 [+0.09, +0.54]). That correlation vanishes once concentration is controlled (partial +0.03 [−0.19, +0.25]), and within the ZZ maps the partial correlations at $\lambda = 0.1$ are −0.31 [−0.56, +0.15] and −0.40 [−0.61, −0.05]. Kernel variance itself tracks the gap better than any g ($\rho = +0.51$ [+0.30, +0.68]). The label-aware ratio $\log_{10}(s_C/s_Q)$ is also negative ($\rho = -0.63$ [−0.75, −0.48]; partial −0.61 [−0.73, −0.44]). But $s_C \geq 4.4$ in every configuration, so the bound $\sqrt{s}$ exceeds 1 and is vacuous. Concentrated kernels get s_Q near 1 because they memorise the labels ($\rho(s_Q, \text{AUC}) = +0.80$ [+0.70, +0.86]).

Huang et al.'s necessary condition. With $N = 500$, $\sqrt{N} = 22.4$. The regularised g never reaches it: the maxima are 19.6 ($\lambda = 0.01$) and 7.4 ($\lambda = 0.1$), so 0 of 79 configurations pass. No configuration shows an advantage, which is consistent with the criterion. The unregularised g (maximum 3.4×10^5) exceeds $\sqrt{N}$

Table 1: Spearman ρ between g and the AUC gap across 79 configurations (95% bootstrap intervals). H2 predicts $\rho > 0$.

	$\lambda = 0$	$\lambda = 0.001$	$\lambda = 0.01$	$\lambda = 0.1$
pooled	-0.50 [-0.65, -0.32]	-0.63 [-0.74, -0.46]	-0.44 [-0.61, -0.22]	$+0.34$ [$+0.09$, $+0.54$]
partial | $\log_{10}$ var.	-0.72 [-0.82, -0.59]	-0.77 [-0.84, -0.67]	-0.66 [-0.75, -0.54]	$+0.03$ [-0.19, $+0.25$]
within Z, $r = 2$	-0.62 [-0.86, -0.29]	-0.79 [-0.92, -0.57]	-0.77 [-0.90, -0.56]	-0.49 [-0.77, -0.10]
within ZZ, $r = 1$	-0.37 [-0.65, -0.01]	-0.43 [-0.69, -0.01]	-0.27 [-0.66, $+0.21$]	$+0.50$ [$+0.07$, $+0.78$]
within ZZ, $r = 2$	-0.25 [-0.55, $+0.12$]	-0.37 [-0.66, $+0.06$]	-0.06 [-0.45, $+0.40$]	$+0.59$ [$+0.23$, $+0.81$]

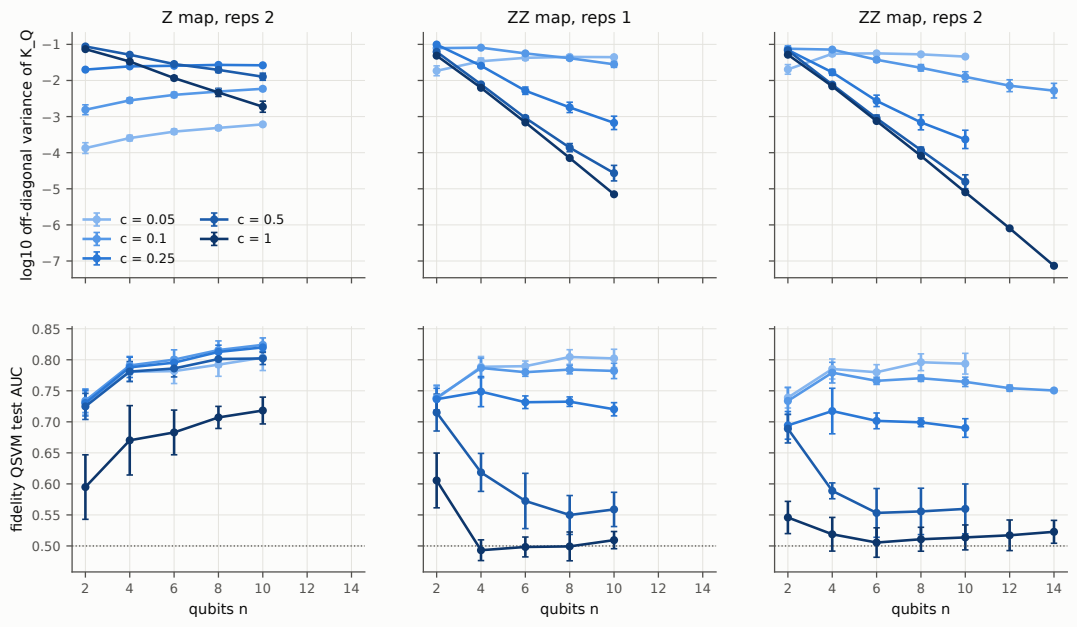


Figure 3: $\log_{10}$ off-diagonal kernel variance (top) and fidelity-kernel AUC (bottom) against qubit count, one line per bandwidth c (mean $\pm$ sd over seeds).

in 61 configurations. None of them shows an advantage either (50 classical better, 11 not significant); these are mostly concentrated kernels, so passing is necessary but not sufficient. **Verdict: H2 failed.** g does not correlate positively with the gap.

5.3 H3: concentration and AUC against qubit count and depth

For each feature map and bandwidth we regress $\log_{10}$ variance and AUC on n (Fig. 3). H3 requires both to fall significantly. That happens in 4 of 15 (map, c) cells: ZZ, reps 1 $c = 0.25$; ZZ, reps 1 $c = 0.5$; ZZ, reps 1 $c = 1$; ZZ, reps 2 $c = 0.5$. For the ZZ maps at $c \geq 0.25$, the variance falls by 0.27–0.49 decades per qubit. For ZZ, $r = 2$, $c = 1$ it drops from 5.2×10^{-2} at $n = 2$ to 7.4×10^{-8} at $n = 14$, while the AUC is already near chance (0.546 and 0.523), so it cannot fall further. For the Z map the AUC rises with n in 5 of 5 cells: more PCA components carry more information. Depth has no consistent effect. Over 25 matched (n, c) cells, ZZ with $r = 2$ is more concentrated than $r = 1$ in 12 and has lower AUC in 14.

Low variance is not enough to predict failure. Concentrated kernels (variance $< 10^{-3}$, off-diagonal mean < 0.1; 17 configurations) have median AUC 0.550 and at most 0.720. Near-constant kernels (variance $< 10^{-3}$, mean > 0.5; 5 configurations, all Z map with the smallest c) still reach median AUC 0.782 in exact simulation. **Verdict: H3 held in part.** It held for the entangling map at larger bandwidths, not for the product map or small c, and not for depth.

5.4 H4: bandwidth at fixed qubit count

In each of the 15 (feature map, n) cells of the main grid we compare $c = 0.05$ with $c = 1$, paired by seed (Fig. 4). *First half.* A smaller c recovers the AUC in 15 of 15 cells; for example, ZZ with $r = 2$ at $n = 8$ goes from 0.511 to 0.796. A smaller c raises the variance (reduces concentration) in only 8 of 15. In the Z-map cells and at $n = 2$, the kernel flattens towards a constant at small c, so its variance falls even though the AUC recovers. *Second half, and its dependence on λ.* For the headline regularised g ($\lambda = 0.01$), g falls steadily towards 1 as

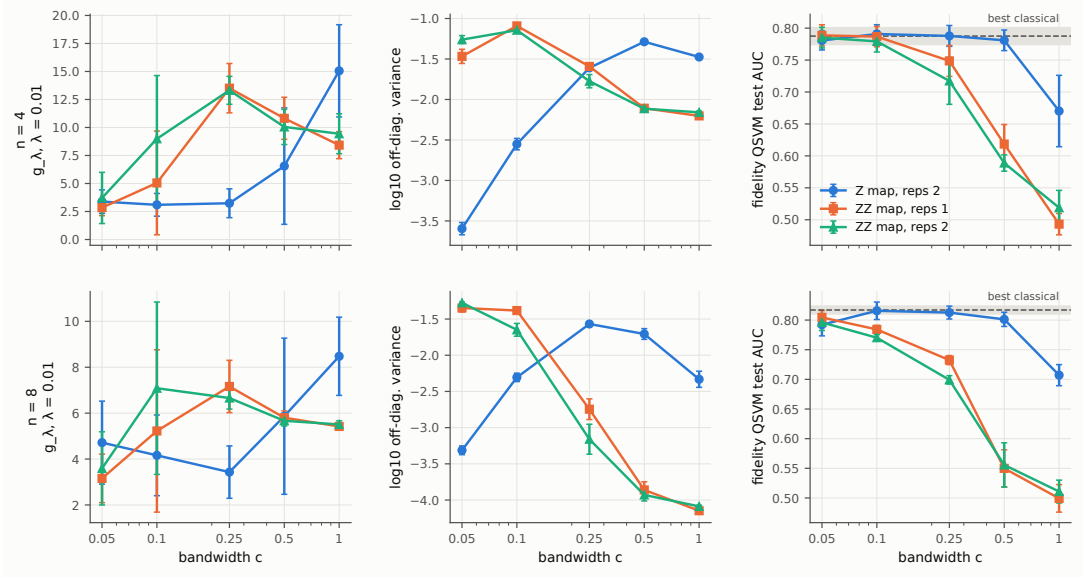


Figure 4: Regularised g ($\lambda = 0.01$), $\log_{10}$ off-diagonal variance and fidelity-kernel AUC against bandwidth c at $n = 4$ (top) and $n = 8$ (bottom); the grey band is the best classical AUC ± sd.

c shrinks in only 7 of 15 cells. For the ZZ maps it peaks at an intermediate bandwidth ($c \in \{0.1, 0.25, 0.5\}$), and at $c = 0.05$ it is still 2.1–4.7. For ZZ, $r = 2$, $n = 8$ it goes 5.5 at $c = 1$, maximum at $c = 0.10$, then 3.6 at $c = 0.05$. The unregularised g ($\lambda = 0$) behaves differently. It is closer to 1 at small c in 15 of 15 cells and falls steadily in 15, from 173–3.4×10^5 at $c = 1$ to 2.8–398 at $c = 0.05$. It moves towards 1 without getting there. **Verdict: H4 held in part.** The first half held for AUC and, for the ZZ maps at $n \geq 4$, for concentration. The second half failed for the regularised g. For the unregularised g only its direction holds.

6 Robustness: PCA versus boosted-tree feature selection

We repeat the $n \in \{4, 8\}$ slices with the top-n features by XGBoost importance, fitted on training events only; `DER_mass_transverse_met_lep` and `DER_mass_vis` are always selected first. All four hypotheses are re-tested on both slices with the same code, and 4 of 4 verdicts are unchanged (Fig. 5).

- **Both sides improve.** Feature selection helps classical and quantum models alike: the best classical AUC moves from 0.788 to 0.825 ($n = 4$) and from 0.817 to 0.831 ($n = 8$). The median change over configurations is +0.026 for the best classical and +0.028 for the fidelity kernel.
- **H1 still holds.** No quantum kernel exceeds the best classical model (0 and 0 of 30). It also holds against the classical model selected by CV, stored only for these runs (0 and 0). Choosing the best classical model on test AUC raises it by only 0.0016 on average.
- **H2 still fails.** ρ is −0.16 [−0.46, +0.21] at $\lambda = 0.01$ (PCA slice: −0.62 [−0.81, −0.30]) and −0.73 [−0.84, −0.54] at $\lambda = 0$. The positive value at $\lambda = 0.1$ (+0.42 [+0.03, +0.69]) again disappears once concentration is controlled (−0.16 [−0.47, +0.24]).
- **H3 and H4 still hold only in part.** H3 holds in 2 of 15 cells (PCA: 1). For H4, a smaller c recovers the AUC in 6 of 6 cells and raises the variance in 4. The regularised g falls steadily towards 1 in 2.
- **Similar angle spread.** The median interquartile range of the encoded angles is 1.31/1.52 rad with feature selection and 1.26/1.01 rad with PCA ($n = 4/n = 8$), so the effective bandwidth is not very different.

7 Noise study and the real-hardware run

Analytic noise. We take four ZZ, $r = 2$ settings: healthy ($c = 0.1$) and concentrated ($c = 1.0$), each at $n = 4$ and $n = 8$. We re-estimate both train and test kernels with binomial shot noise (100, 1000, 10000 shots) and global depolarising noise ($p_{\text{tot}} \in \{0.01, 0.05, 0.1\}$), 5 seeds each (Fig. 6a,b). Shot noise dominates. The healthy $n = 4$ kernel goes from AUC 0.779 (exact) to 0.748 (1000 shots) and 0.706 (100 shots). The $n = 8$

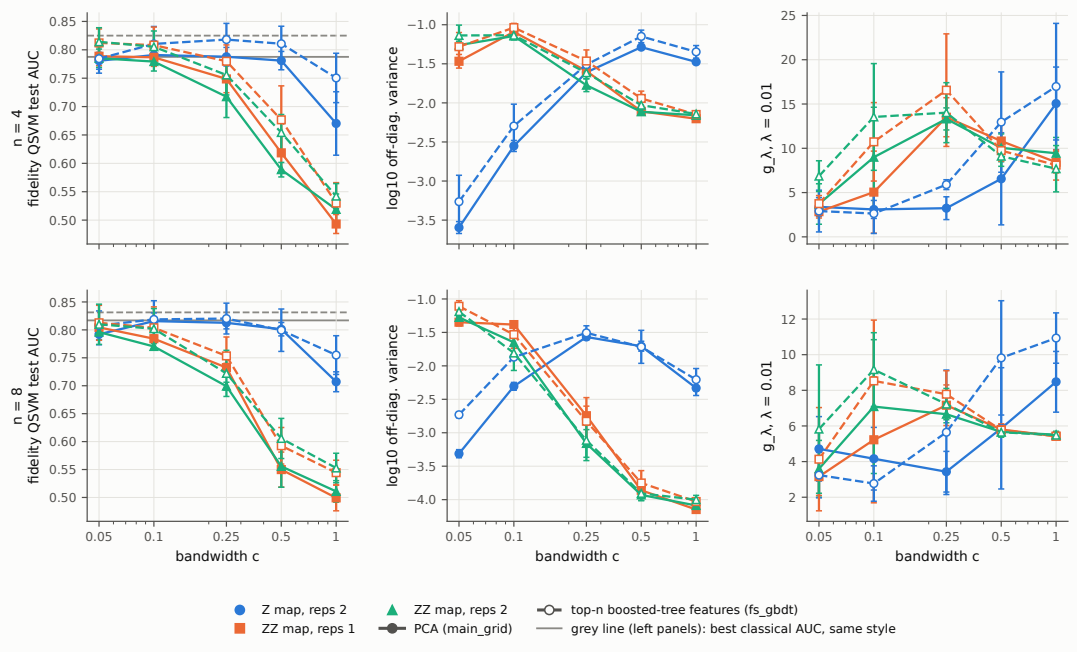


Figure 5: Fidelity-kernel AUC, $\log_{10}$ variance and g ($\lambda = 0.01$) against c at $n = 4$ and $n = 8$, for PCA (solid) and the top-n boosted-tree features (dashed); grey lines are the best classical AUC.

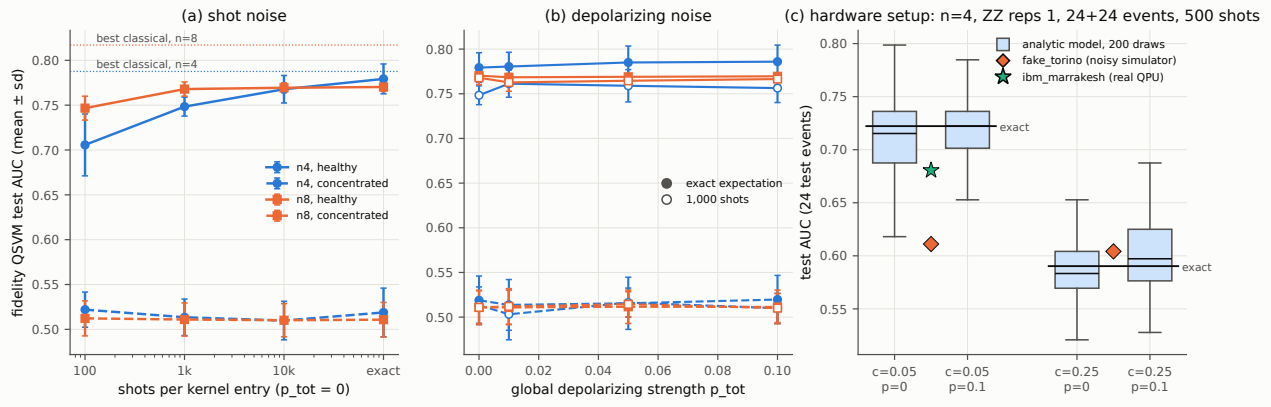


Figure 6: (a) Fidelity-kernel AUC against shots per kernel entry; (b) against global depolarising strength; (c) the hardware setup: analytic noise model (boxes over 200 draws), noisy simulator and `ibm_marrakesh`.

kernel goes from 0.770 to 0.747. Depolarising noise alone barely matters: 0.786 at $p_{\text{tot}} = 0.1$ for healthy $n = 4$. The concentrated kernels stay at 0.503–0.524 under every condition, against best classical AUCs of 0.788 and 0.817. Shot noise also inflates the measured variance of the concentrated $n = 8$ kernel, from 8.18×10^{-5} to 1.35×10^{-4} at 100 shots. So the variance diagnostic is reliable at 1000 or more shots but can mislead at 100.

Hardware. On `ibm_marrakesh` we ran the ZZ map with $r = 1$, $n = 4$, $c = 0.05$, with 24 training and 24 test events and 500 shots. This took 996 circuits (mean depth 41, 9 two-qubit gates) and 143 s of QPU time (Fig. 6c).

- **Fidelity kernel:** AUC 0.681, against 0.722 exact and 0.611 on the `fake_torino` noisy simulator.
- **Projected kernel:** 0.743, against 0.806 exact and 0.771 simulated.
- **Kernel error:** the mean absolute error of the hardware training kernel is 0.041 (simulator: 0.042). Its off-diagonal mean drops from 0.528 to 0.489, which corresponds to an effective global depolarising strength of 0.082 (simulator: 0.090).
- **Analytic model:** at $p_{\text{tot}} = 0.1$ and 500 shots, the model gives a mean AUC of 0.718 (5–95%: 0.645–0.785) and a kernel error of 0.048. The hardware AUC falls inside that range; the simulator's falls below it.

With only 24 test events, these AUCs are demonstrations, not precise measurements.

8 Discussion and limitations

What the results say about g. Huang et al. use g as a *necessary condition*. A small g guarantees that a classical model is competitive; a large g only opens the possibility of an advantage, which must then be confirmed with the label-aware complexities [13]. Our data are consistent with that: the regularised g never passes $\sqrt{N}$, and no configuration beats the classical baselines. What fails is the *graded* reading of g, in which a larger g marks a more promising dataset or configuration. On these data g is anti-correlated with the gap, and its only positive association disappears once kernel concentration is controlled. Concentration explains the results better: it tracks the gap more strongly than any g. It accounts for the ZZ collapse at large c and n. It also shows two distinct low-variance regimes, a concentrated one and a near-constant one. This matches the "spectral degeneracy" confound reported for power-system data [14] and the bandwidth-to-RBF picture [11, 21]. A plausible reason why the regularised g does not return to 1 at small c is that the small-bandwidth ZZ kernel approaches an *anisotropic* Gaussian [22], not the isotropic RBF used as K_C. We have not tested this.

Limitations.

- **Simulated data and small samples.** The events are detector-level Monte Carlo, with balanced classes and only $N = 500$ training events, far fewer than the classical models could use.
- **Lenient significance.** 5 seeds and the 1-sd rule correspond to $p \approx 0.09$ per comparison, with many comparisons. "Not significant" does not mean equal.
- **Shared seeds.** All configurations reuse the same 5 data splits, so they are not independent, and the bootstrap intervals over configurations are optimistic.
- **Best-of-three classical.** "Best classical" is selected on test AUC. This favours H1. It was measured to be small (0.0016), but only for the feature-selection runs.
- **Hardware.** Only one configuration was run on hardware, with 24 test events. The noise model is a global depolarising channel, not a device model.
- **Scope.** Only Z and ZZ maps with linear entanglement, one classical reference kernel for g, and a headline λ that is our choice, not part of the hypotheses. The hypotheses were written before the experiments but not formally pre-registered.

9 Conclusion and future work

On ATLAS $H \to \tau\tau$ events, tuned classical baselines match or beat every quantum kernel we test (H1 held), so we claim no quantum advantage. The geometric difference fails as a graded predictor of the quantum-minus-classical gap (H2 failed). It is anti-correlated with the gap, consistent with Huang et al.'s necessary condition, and better explained by kernel concentration. Concentration rises with qubits at fixed bandwidth for the entangling map but not in general (H3 held in part). A smaller bandwidth recovers AUC but does not pull g back towards 1 (H4 held in part). These conclusions survive a change of feature reduction and the noise levels we studied.

Next steps follow from the literature and are cheap with the stored kernels:

- Kernel–target alignment as the predictor, which worked where g failed on power-system data [14].
- g against an anisotropic Gaussian reference [22].
- Shot-noise requirements for the near-constant small-c kernels [16].
- Larger training sets and more hardware test events.

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